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Hybrid MoM-PO Solver for Antenna Placement Analysis

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Problem Statement

- The main problem addressed in this project is the development of a scattering solver capable of accurately calculating the scattered magnetic field when a known incident magnetic field interacts with an object.
- The challenge is to combine the MoM for the antenna's fine details and PO for the electrically large structure upon which it is placed, ensuring computational efficiency and accuracy.

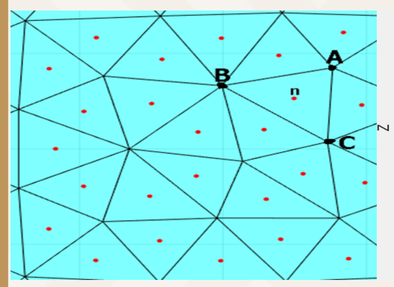
Project Aim

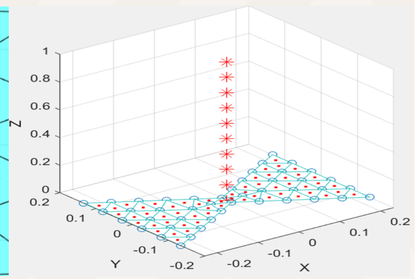
- The aim of this project is to develop a scattering solver that uses Physical Optics (PO) and the Method of Moments (MoM) to accurately calculate the magnetic field radiated by an antenna positioned close to a large structure.

Objectives

- Formulate the scattered problem using integral equations.
- Implement PO and MoM solvers for plane wave scattering in MATLAB.
- Develop a Hybrid MoM-PO solver in MATLAB to solve the radiated magnetic field.
- Validate the solver with known solutions and experimental data using MATLAB & FEKO.
- Verify near-field and far-field patterns by comparing MATLAB with FEKO.

Discretization of the current



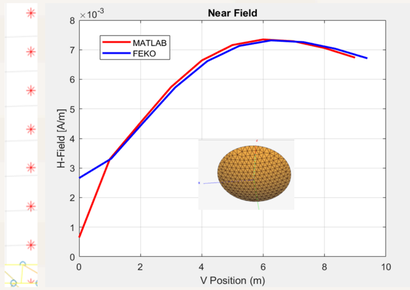


- Create a mesh with triangular elements.
- Calculate the surface current density at the centroid of the triangles.
- The centroid coordinates at each triangle are given by this equation:

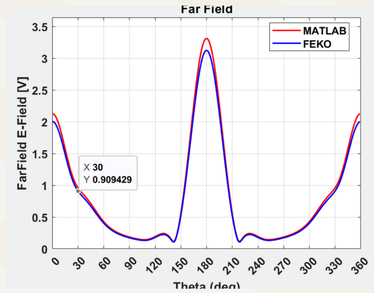
$$centroid\,(x, y, z) = \frac{(A + B + C)}{3}$$

Results: Physical Optics (PO) Solver

- The near-field pattern |H_s(r)| vs 10 line of points



- The far-field pattern $\phi = 0$ and $\theta = [0 - 360]$

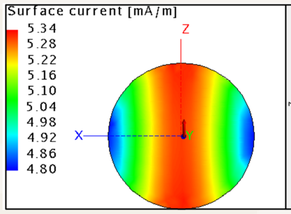


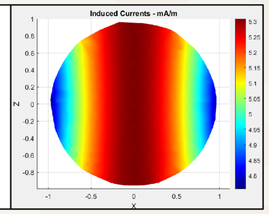
Results: Physical Optics (PO) Solver

Angle θ and φ	FEKO		MATLAB	
	$\lvert J_{PO}\rvert$ [mA/m]	$\lvert H_s\rvert$ [mA/m]	$\lvert J_{PO}\rvert$ [mA/m]	$\lvert H_s\rvert$ [mA/m]
0°	5.3401	7.153	5.309	6.853
30°	5.400	2.065	5.3089	1.891
45°	5.391	3.281	5.391	3.133
60°	5.392	3.367	5.391	3.321
90°	5.401	3.601	5.384	3.582

- Comparison of the magnitude of $|J_{PO}|$ and H-Field in (mA/m) at various angles using FEKO and MATLAB simulations.

- Figure showing the current distribution of the surface current density for MATLAB and FEKO for the PO $\lambda \times \lambda$ parabolic plate.



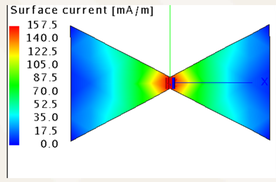


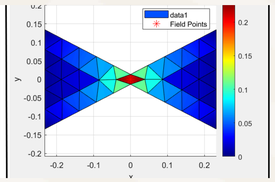
Results: Method of Moments (MoM) Solver

Excitation Voltage [V]	FEKO		MATLAB	
	$\lvert J_{MoM}\rvert$ [mA/m]	$\lvert H_s\rvert$ [mA/m]	$\lvert J_{MoM}\rvert$ [mA/m]	$\lvert H_s\rvert$ [mA/m]
$V = 1.0\angle 0$	157.5	112.5	156.41	112.04
$V = 0.8\angle 0$	150.0	90.0	147.3	89.63
$V = 0.6\angle 0$	112.5	75.0	109.7	73.2
$V = 0.4\angle 0$	75.0	45.0	67.4	44.81
$V = 0.2\angle 0$	32.0	22.5	31.28	22.4
$V = 0.0\angle 0$	0	0	0	0

- Comparison of the magnitude of $|J_{MoM}|$ and H-Field in (mA/m) at various angles using FEKO and MATLAB simulations.

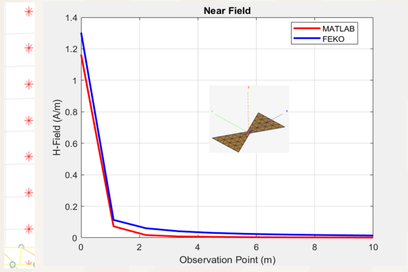
- Figure showing the current distribution of the surface current density for MATLAB and FEKO for the MoM bowtie antenna





Results: Method of Moments (MoM) Solver

- The near-field pattern |H_s(r)| vs 10 line of points



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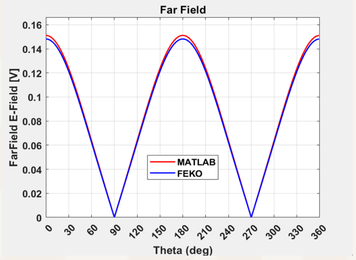
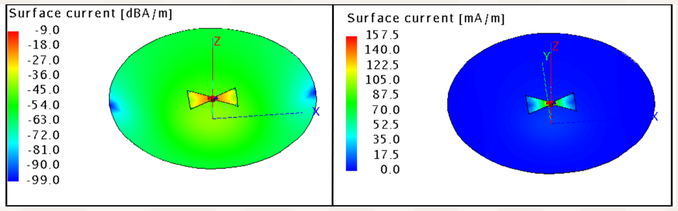
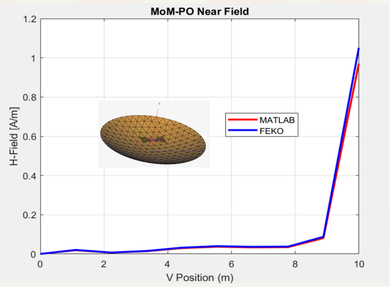
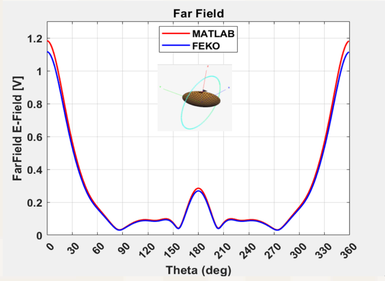


Results: Hybrid MoM-PO Solver







Conclusion

- Developed and implemented PO, MoM, and Hybrid MoM-PO solvers in MATLAB to accurately analyze plane wave scattering and magnetic fields.
- Verified the solvers by comparing MATLAB results with experimental data from FEKO, showing good agreement in both near-field and far-field.
- Proved that these methods are reliable for modeling complex electromagnetic interactions, helping in antenna and reflector design.